

Dataset Report — Understanding the Three Benchmark Network-Intrusion Datasets

Project: Evaluating and Improving Cross-Dataset Generalization of ML-Based Intrusion Detection Systems **Document:** 01 — Dataset Analysis **Status:** Prepared for mid-semester review

1. Executive Summary

This report documents a first-hand analysis of the three benchmark network-traffic datasets selected for this project: **CIC-IDS-2017**, **UNSW-NB15**, and **TON_IoT**. The analysis was performed directly on the downloaded CSV files (statistics recomputed by the authors, not quoted from the literature), and it reveals the single most important structural fact of the project:

The three datasets were produced by **different collection tools and represent almost disjoint feature spaces**. CIC-IDS-2017 exposes 78 CICFlowMeter flow statistics, UNSW-NB15 exposes 42 Argus/Bro hybrid counters, and TON_IoT exposes 42 raw Zeek (Bro) log fields. Only a small handful of features are semantically equivalent across all three. Any “cross-dataset” experiment must therefore either (a) restrict itself to a small shared feature subspace, or (b) build dataset-specific models that cannot be directly transferred.

This feature-schema mismatch is not an obstacle to be worked around — it is the central phenomenon the project studies, because it is precisely what a real, deployable Intrusion Detection System (IDS) faces when it encounters a network it has never seen before.

All figures referenced in this document are stored under `reports/figures/`.

2. Datasets at a Glance

Property	CIC-IDS-2017	UNSW-NB15	TON_IoT (network)
Origin	Canadian Institute for Cybersecurity (CIC), Univ. of New Brunswick	Australian Centre for Cyber Security (ACCS), UNSW Canberra	UNSW Canberra / Moustafa et al.
Year	2017	2015	2019–2020
Collection tool	CICFlowMeter (flow exporter)	Argus + Bro-IDS hybrid	Zeek (Bro-IDS) connection logs
Network type	Enterprise/office LAN	Emulated enterprise (IXIA PerfectStorm)	IoT / IIoT testbed (edge–fog–cloud)
Total records (this copy)	2,830,743	257,673 (train 175,341 + test 82,332)	211,043
Feature columns	78 (one duplicated → 77 unique)	42 (+ id, attack_cat, label)	42 (+ label, type)
Label scheme	Label (per-attack string)	attack_cat (9 classes + Normal) + binary label	type (10 classes) + binary label
Missing / infinite values	1,358 NaN + 4,376 Inf	none	none
Key structural traits	Extreme class imbalance; identity features (ports)	Partly de-identified (IPs/ports already removed)	Heavy categoricals (conn_state, dns_*, http_*, ssl_*)

3. CIC-IDS-2017

3.1 Provenance and collection methodology

CIC-IDS-2017 [5] is the most widely used intrusion-detection benchmark. It was captured at the Canadian Institute for Cybersecurity over five working days (Monday–Friday, July 2017) on a realistic testbed that emulated a small enterprise network: victim machines, a firewall, and an attacker network. Realistic background (benign) traffic was generated with a behavioural profiling tool (B-Profile), while attacks were executed on specific days. Flows were extracted with **CICFlowMeter**, which computes 78+ statistical features per bidirectional flow (packet-length statistics, inter-arrival times, TCP flag counts, bulk and sub-flow statistics, and active/idle timing).

The dataset ships as eight per-day CSV files under `datasets/CIC-IDS-2017/MachineLearningCVE/`.

3.2 File-by-file composition (measured)

File	Rows	Attack labels present
Monday-WorkingHours	529,918	BENIGN only
Tuesday-WorkingHours	445,909	BENIGN, FTP-Patator, SSH-Patator
Wednesday-workingHours	692,703	BENIGN, DoS Hulk/Golden-Eye/slowloris/Slowhttptest, Heartbleed
Thursday-WorkingHours-Morning-WebAttacks	170,366	BENIGN, Web Attack – Brute Force/XSS/Sql Injection
Thursday-WorkingHours-Afternoon-Infiltration	288,602	BENIGN, Infiltration
Friday-WorkingHours-Morning	191,033	BENIGN, Bot
Friday-WorkingHours-Afternoon-DDoS	225,745	BENIGN, DDoS
Friday-WorkingHours-Afternoon-PortScan	286,467	BENIGN, PortScan

Full label distribution (log scale):

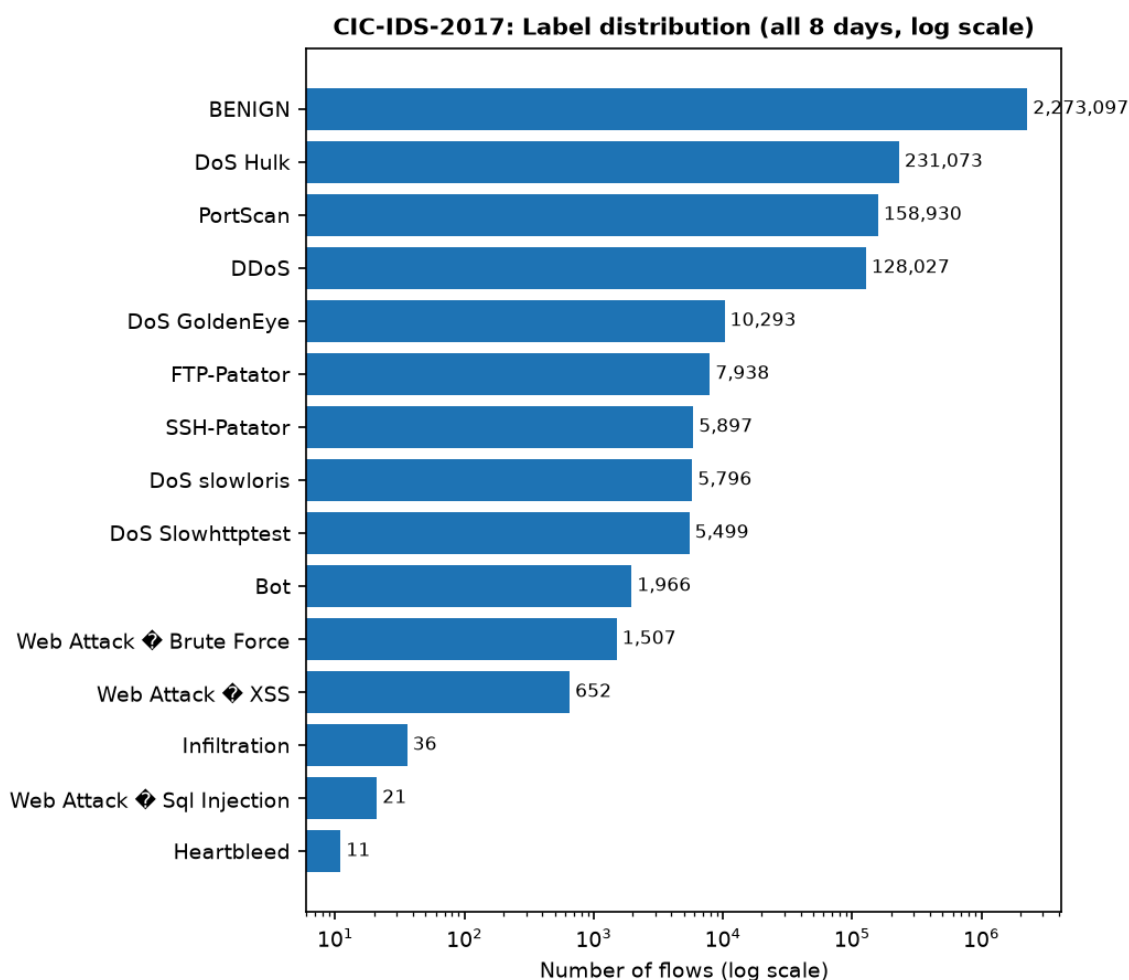


Figure 1: CIC-IDS-2017 class distribution

Key observations:

- **BENIGN dominates** (2,273,097 of 2,830,743 $\approx$ 80.3%). Attacks total $\approx$ 557,646 (19.7%).
- **DoS Hulk is the single largest attack class** (231,073), followed by PortScan (158,930) and DDoS (128,027).
- **Extreme minority classes:** Heartbleed (11) and Infiltration (36) are effectively unlearnable in a per-dataset setting and would be *impossible* to transfer cross-dataset.
- **Label encoding issue:** the Web-Attack labels contain a non-ASCII en-dash that reads as a replacement character (Web Attack ⚡ Brute Force) under default UTF-8 decoding. Labels must be normalised before use.

3.3 Feature schema

The header contains **79 columns**: 78 numeric feature columns plus the Label column. One feature — Fwd Header Length — is **duplicated** (appears twice; pandas renames the second occurrence to Fwd Header Length.1), so there are **77 unique features**. Notable schema characteristics:

- Features are **numeric flow statistics** (no categorical protocol/service columns).
- The only categorical column is `Label`.
- The first column `Destination Port` is a *de-facto identity feature* (see §6): it encodes the attack scenario rather than traffic behaviour and is a known source of inflated same-dataset accuracy [1], [15].

3.4 Data-quality issues (measured)

- **Infinite values:** 4,376 cells equal Infinity, concentrated in `Flow Bytes/s` and `Flow Packets/s` (a division-by-zero artefact when a flow has zero packets). These must be replaced (e.g. with `NaN/median/zero`) before training tree models that reject infinities.
- **Missing values:** 1,358 cells are `NaN` (same columns, plus a few scattered others).
- Combined, non-finite values are a tiny fraction of the ~221 million feature cells (~0.0026%), but they are exactly the rows/columns a robust pipeline must handle.
- **Imbalance** (see §3.2) means per-class metrics (not overall accuracy) are mandatory.

3.5 Implications

CIC-IDS-2017 provides rich flow statistics, but (i) its port/identity features, (ii) its extreme imbalance, and (iii) its non-finite values make it the dataset most prone to giving a *falsely high* same-dataset score that fails to transfer. This is exactly the behaviour documented in [1].

4. UNSW-NB15

4.1 Provenance and collection methodology

UNSW-NB15 [6] was produced at the ACCS (UNSW Canberra) in 2015 using the **IXIA PerfectStorm** traffic generator to emulate nine modern attack families (Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms) alongside realistic background traffic. Features were extracted with the **Argus** tool augmented by **Bro-IDS** scripts, yielding a hybrid set of flow counters, connection-state counters, and time-window-based connection-count features (`ct_*`).

This copy ships as `UNSW_NB15_training-set.csv` (175,341 rows) and `UNSW_NB15_testing-set.csv` (82,332 rows) under `datasets/UNSW-NB15/`.

4.2 Composition (measured)

Set	Nor- mal	Generic	Ex- ploits	Fuzzers	DoS	Re- con	Analy- sis	Back- door	Shell- code	Worms
Train	56,000	40,000	33,393	18,184	12,264	10,491	2,000	1,746	1,133	130
Test	37,000	18,871	11,132	6,062	4,089	3,496	677	583	378	44
Total	93,000	58,871	44,525	24,246	16,353	13,987	2,677	2,329	1,511	174

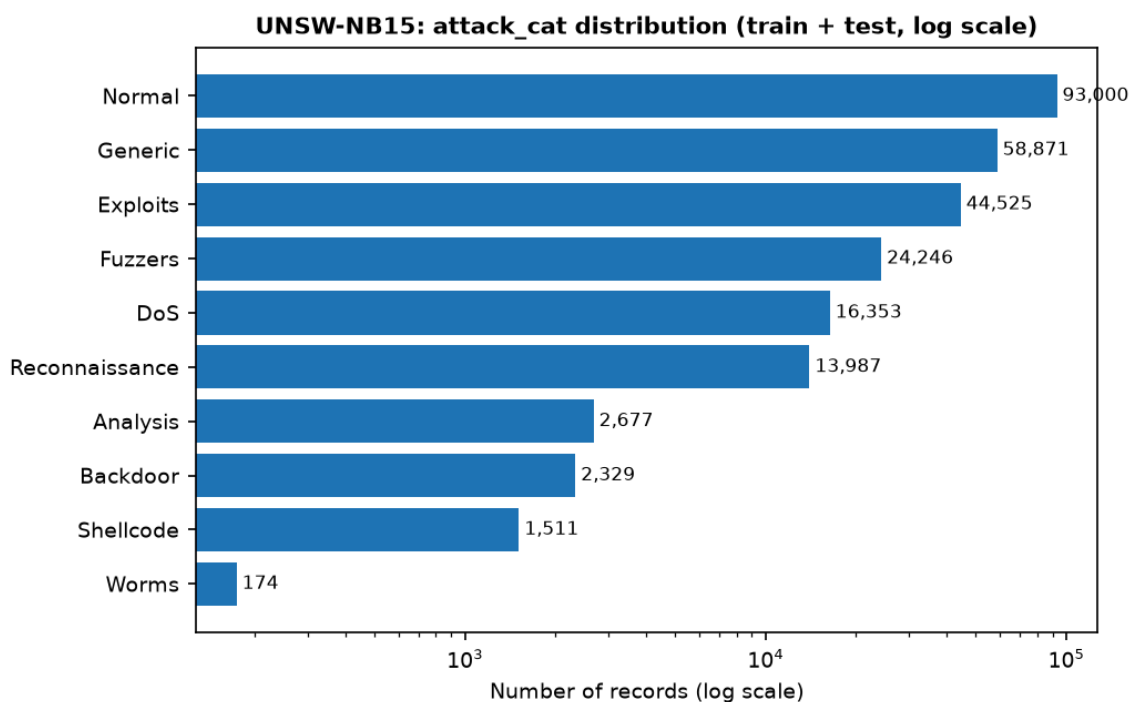


Figure 2: UNSW-NB15 attack distribution

Key observations:

- Binary split is well balanced: `label=1` (attack) = 164,673 vs `label=0` (Normal) = 93,000.
- `attack_cat` is imbalanced across the 9 attack classes (Worms = 174 vs Generic = 58,871).
- The provided CSVs already **exclude** the raw identity columns (`srcip`, `sport`, `dstip`, `dsport`) that are present in the full 49-column canonical release — this copy is the reduced 45-column variant (`id` + 42 features + `attack_cat` + `label`).

4.3 Feature schema

42 features, a mix of numeric counters and a few categoricals (`proto`, `service`, `state`). The `ct_*` features (e.g. `ct_srv_src`, `ct_dst_ltm`) count how many of the last 100 connections share the same service/host/port — these are **temporal-window aggregations that depend on the collection window** and are a known source of non-transferable signal [1], [3]. Two timestamp columns (`stime`, `ltime`) are absent in this variant.

4.4 Data quality

No missing or infinite values were found. The main caveats are (i) the train/test split is a provided fixed split rather than a random partition, and (ii) the `ct_*` features are time-dependent and leak collection-order information.

5. TON_IoT (network)

5.1 Provenance and collection methodology

TON_IoT [7] is a heterogeneous IoT/IIoT dataset collected from a three-layer testbed (edge, fog, cloud) orchestrated with SDN (VMware NSX) and NFV. The network portion contains traffic from IoT sensors and hosts, with attacks in ten families. Network features are the **raw Zeek (Bro) connection-log fields** — a fundamentally different feature philosophy from CICFlowMeter (no aggregate statistics; instead protocol-specific fields such as `dns_*`, `http_*`, `ssl_*`, and `conn_state`).

This copy ships as the combined `train_test_network.csv` (211,043 rows) under `datasets/TON_IOT network train-test/Train_Test_Network_dataset/`, plus 24 per-day `Network_dataset_*.csv` files under `datasets/TON_IOT network processed/`.

5.2 Composition (measured)

Class	normal	backdoor	ddos	dos	injection	pass-ransomwarescan-word	scanning	xss	mitm
Count	50,000	20,000	20,000	20,000	20,000	20,000	20,000	20,000	1,043

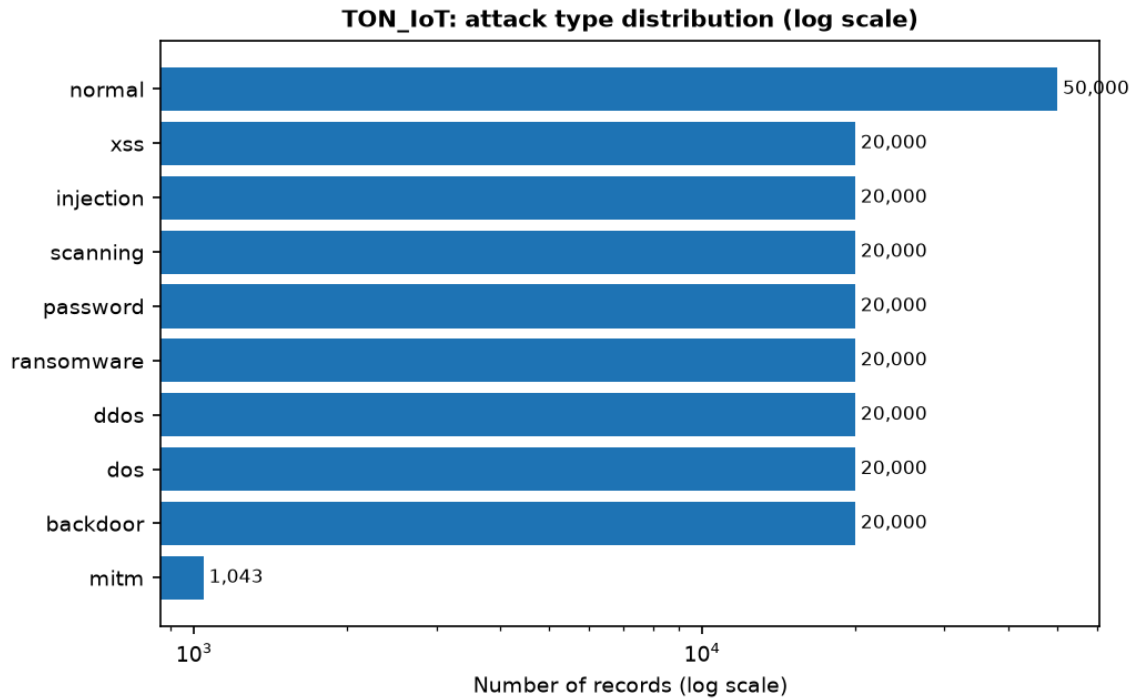


Figure 3: TON_IoT type distribution

Key observations:

- Attack classes are **near-uniformly balanced** (20,000 each) except mitm (1,043) — a deliberate design choice that removes the class-imbalance confound present in the other two datasets.

- Binary split: label=1 (attack) = 161,043 vs label=0 (normal) = 50,000.

5.3 Feature schema

42 feature columns plus label and type. Unlike the other two datasets, **most columns are categorical strings**: proto, service, conn_state, dns_query, dns_*, ssl_*, http_method, http_uri, http_user_agent, weird_*, etc. Only ~15 columns are numeric (duration, src_bytes, dst_bytes, src_pkts, dst_pkts, src_ip_bytes, dst_ip_bytes, port fields, and a few counters).

Implications:

- Requires **one-hot / target encoding** of high-cardinality categoricals (http_uri, dns_query have very large vocabularies).
- Contains explicit **identity columns** (src_ip, dst_ip, src_port, dst_port) that must be dropped or treated with extreme care to avoid memorisation.
- The Zeek schema is the “rawest” of the three — closest to what a production sensor emits — which makes it the most realistic *target* domain for transfer.

6. Cross-Dataset Comparison — The Feature-Schema Mismatch

6.1 Feature-space size

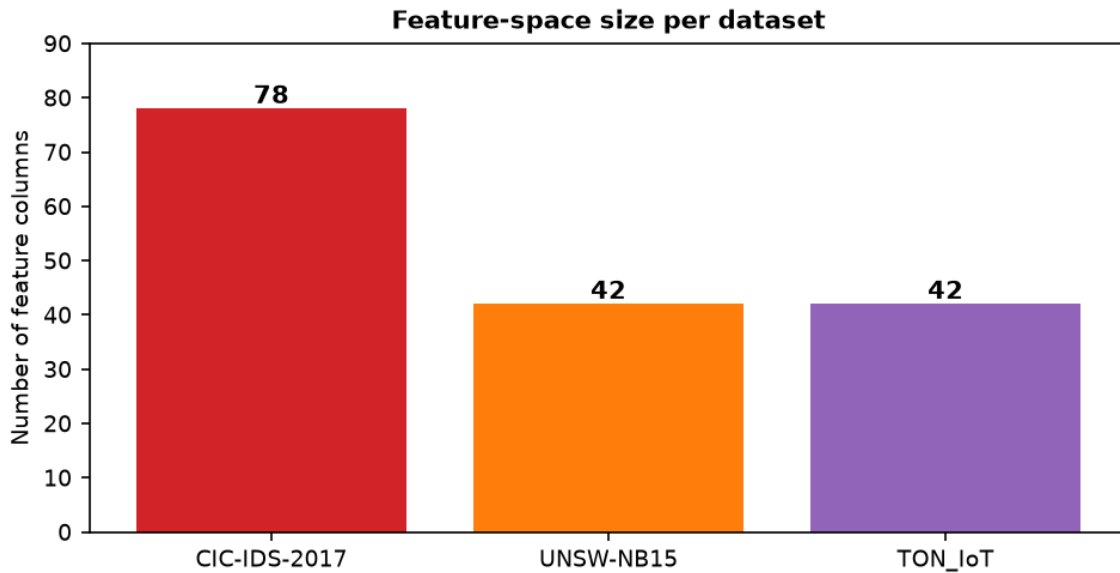


Figure 4: Feature counts per dataset

The three datasets expose comparable numbers of columns (78 vs 42 vs 42), but these columns are **almost disjoint**. There is no shared column name across all three datasets.

6.2 Semantic feature mapping (the shared subspace)

To run any cross-dataset experiment, one must map features to a **common semantic subspace**. The table below is the core of our feature-harmonisation design. A ✓ in three columns means the feature can be constructed in all three datasets directly.

#	Semantic concept	CIC-IDS-2017	UNSW-NB15	TON_IoT	All three?
1	Flow/connection duration	Flow Duration	dur	duration	✓
2	Source→dest bytes	Total Length of Fwd Packets	sbytes	src_bytes	✓
3	Dest→source bytes	Total Length of Bwd Packets	dbytes	dst_bytes	✓
4	Source→dest packets	Total Fwd Packets	spkts	src_pkts	✓
5	Dest→source packets	Total Backward Packets	dpkts	dst_pkts	✓
6	Protocol	(implicit via flags/port)	proto	proto	partial
7	Service	Destination Port (proxy)	service	service	partial
8	Connection state	(TCP flag counts, proxy)	state	conn_state	partial
9	Byte/packet rate	Flow Bytes/s, Flow Packets/s	rate, sload, dload	derivable	partial
10	Packet-size statistics	* Packet Length * (16 cols)	smean, dmean	—	X
11	Inter-arrival/jitter	Flow IATs in Fwd/Bwd IAT *	inpkt,dinpkt,sjit,djit	—	X
12	TCP flags	SYN/FIN/RST/PSH/ACK/URG/ECE/CWE Flag Count	—	—	X
13	TCP window / bulk	Init_Win_bytes * Bulk *, Subflow *	win,dwin,stcpb,dtcpb	—	X
14	UNSW-NB15 connection fields	(partly transacted Subflow)	src_ip,dst_ip,src_port,dst_port	src_ip,dst_ip,src_port,dst_port	partial
15	UNSW-NB15 connection fields	(partly transacted Subflow)	src_ip,dst_ip,src_port,dst_port	src_ip,dst_ip,src_port,dst_port	partial

Conclusion: the *directly shared* numeric subspace is roughly **6 raw features** (duration, src/dst bytes, src/dst packets, plus proto/service/state after encoding). After one-hot encoding protocol/service/state and deriving rate-like ratios, this expands to roughly **15–25 comparable dimensions**. This is consistent with prior cross-dataset work that deliberately restricts to a common representation [1], [3], [4] — and it is the feature set our metaheuristic optimiser will search over.

6.3 Class-taxonomy mapping (binary + coarse multi-class)

Because each dataset names attacks differently, cross-dataset *multi-class* evaluation requires a shared taxonomy. We propose the following mapping (binary = benign vs attack is the primary target; coarse multi-class is the secondary target):

Coarse class	CIC-IDS-2017		UNSW-NB15	TON_IoT
Benign / Normal	BENIGN		Normal	normal
DoS / DDoS	DoS	Hulk, GoldenEye, slowloris, Slowhttptest, DDoS	DoS	dos, ddos
Reconnaissance / Scanning	PortScan		Reconnaissance, Analysis	scanning
Brute-force / Credential	FTP-Patator, Patator	SSH-	(part of Generic)*	password
Backdoor / Bot	Bot		Backdoor	backdoor
Web / Injection	Web Brute Sql	Attack Force/XSS/ Injection	Exploits, Shellcode*	injection, xss
Generic / Exploit / Worm	—		Generic, Exploits, Shellcode, Fuzzers, Worms,	—
Ransomware	—		—	ransomware
MITM	—		—	mitm
Other (rare)	Heartbleed, Infiltration		—	—

* Note: the UNSW Generic/Exploits/Shellcode categories do not cleanly map to the others; this *semantic fuzziness* is itself a finding and a documented cause of the generalisation gap (Objective 2 of the project).

7. What This Means for the Project

1. **Feature harmonisation is unavoidable and is a first-class deliverable**, not a preprocessing detail.
 2. **Identity/leakage features must be removed** (CICIDS Destination Port; TON src_ip, dst_ip, src_port, dst_port) — otherwise cross-dataset scores measure memorisation.
 3. **Non-finite and categorical handling differs per dataset**, so the pipeline must be dataset-aware yet produce a *common* numeric matrix.
 4. **Class imbalance is dataset-specific** (CICIDS extreme, TON balanced), so evaluation must use per-class and macro metrics, not overall accuracy.
 5. The **shared subspace is small** (~15–25 dims), which makes feature *selection* (the focus of our metaheuristic work) both more important and more tractable.
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References

Reference numbers follow the canonical 15-paper list in 02_literature_review.md and 05_midsem_report.md.

- [1] M. Cantone, C. Marrocco, and A. Bria, “On the cross-dataset generalization of machine learning for network intrusion detection,” *IEEE Access*, vol. 12, pp. 144489–144508, 2024.
- [3] M. A. Hakim, M. S. Uddin, and T. I. Anis, “Cross-domain generalization failure in lightweight intrusion detection models for IIoT networks,” arXiv:2607.00553, 2026.
- [4] M. U. Butt, A. Hotho, and D. Schlör, “Evaluating tabular representation learning for network intrusion detection,” in *Proc. IEEE CSR*, 2026.
- [5] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward generating a new intrusion detection dataset and intrusion traffic characterization,” in *Proc. ICISSP*, 2018.
- [6] N. Moustafa and J. Slay, “UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set),” in *Proc. MilCIS*, 2015.
- [7] N. Moustafa, “A new distributed architecture for evaluating AI-based security systems at the edge: Network TON_IoT datasets,” *Sustain. Cities Soc.*, vol. 72, 2021.
- [15] J. Vitorino, M. Silva, E. Maia, and I. Praça, “An adversarial robustness benchmark for enterprise network intrusion detection,” in *Proc. FPS*, 2023.